Report 1
Group 6

Internet use and loneliness

Research question

Is people's internet use associated with their feeling of loneliness?

Findings

Among regular internet users in Switzerland, those above the mean in self-reported time online report no more loneliness than those at or below it.

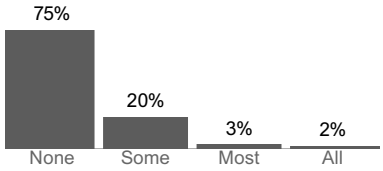
European Social Survey Round 11
Switzerland, 2023
N 1,384 respondents

1 Variables

Dependent
`fltlnl`

For our dependent variable, we chose `fltlnl` ("Felt lonely, how often past week", 1–4) as a proxy for the self-reported loneliness of our respondents. In ESS 11, it is the only item about the feeling of loneliness itself. Others like `fltdpr` and `fltsd` report bad feelings like depression or sadness, but no others report loneliness. The main limitation is the answer distribution of this variable (Figure 1). The vast majority report no loneliness, which leads to very little variation to explain and group differences being small.

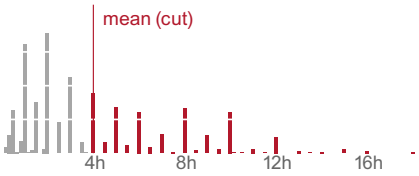
Figure 1 Self-reported loneliness, past week



Independent
`netustm`

`netustm` ("Internet use, how much time on typical day, in minutes") was chosen as the independent variable. While there are other variables like `netusoft` also reporting internet usage, `netustm` is continuous and further spread out. `netusoft` is coarser and four in five respondents answered with "every day". The answer distribution of `netustm` is shown in Figure 2. A major limitation is that this question was only asked of people who had previously reported using the internet most days or every day. This led to 168 respondents being skipped and never shown this question. Additionally, the reported time also includes work use.

Figure 2 Self-reported minutes online per day



2 Grouping

`net_group`

In order to perform a t-test later on, a new grouping variable `net_group` was created, which splits `netustm` at its mean of 235.6 minutes. 1 is assigned to values above the mean, 0 for values exactly at or below the mean. No respondent sits exactly on the mean, and `netustm` itself is left unchanged. Because the mean of `netustm` lies above its median, fewer answers fall above it, which leads to groups of unequal size (Figure 3).

Figure 3 Groups, N 1,212, mode 0



3 Descriptives

After recoding the text labels in the CSV to numbers (Table 1), for `fltlnl` median, mode, and IQR are all at the bottom of the scale, while for `netustm` the mean is far above the median (235.6 against 180). Looking at its distribution, it is strongly right-skewed and answers also heap at whole hours, but the split at the mean still produced two sufficiently large groups (Figure 3).

Table 1	`fltlnl`	`netustm`
Valid N	1,382	1,212
Mean	1.31	235.6
Median	1	180
Mode	1	120
Variance	0.38	39,098
SD	0.61	197.7
Range	1–4	4–1,080
IQR	0	270

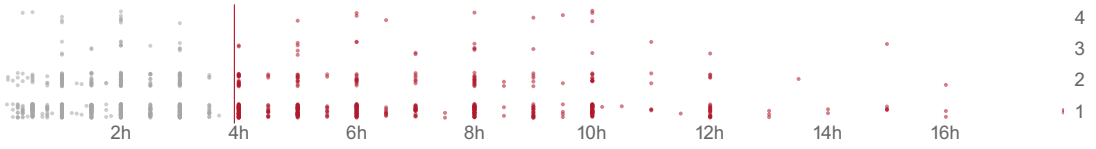
4 Test

We chose to use Welch's two-sample t-test, which is the default of R. This works well because our groups are independent, unequal in size and spread, and Welch does not assume equal variances.

Group	N	SD
0 at or below	715	0.59
1 above	495	0.65

Figure 5

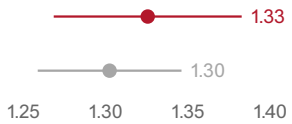
Every respondent, self-reported loneliness (1–4, jittered) by self-reported minutes online, red above the mean



5 Interpretation

To answer our research question, the t-test compares the mean self-reported loneliness of the two groups (Figure 4). Respondents above the mean report slightly more loneliness than those at or below it, but the difference is very small and not statistically significant. For regular internet users in Switzerland, time spent online therefore tells us almost nothing about the reported feeling of loneliness. A limitation is the cut point itself, since splitting at the mean only compares two group averages and ignores all variation within each group. Figure 5 displays all respondents' reported answers. It shows that loneliness is spread similarly across both sides of the cut. In Report 2, the regression will use `netustm` as a continuous variable instead of the split.

Figure 4 Mean self-reported loneliness, 95% CI
Welch 1 − 0: +0.02 (−0.05 to 0.09)
t 0.63, df 1004, p 0.53



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